INFORMATION AND TECHNOLOGY UNIVERSITY LHR

Deep Metric Learning for Image Retrieval

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1.Introduction:

This assignment explores Deep Metric Learning for image retrieval using the **Caltech-101 dataset**. Unlike standard classification, the network learns a compact embedding vector for each image. Images from the same class are close together in the embedding space, while images from different classes are far apart.

Objectives:

1- Train embedding networks using contrastive loss and triplet loss.

2- Implement pair and triplet sampling strategies.

3- Evaluate embeddings using Recall@K metrics.

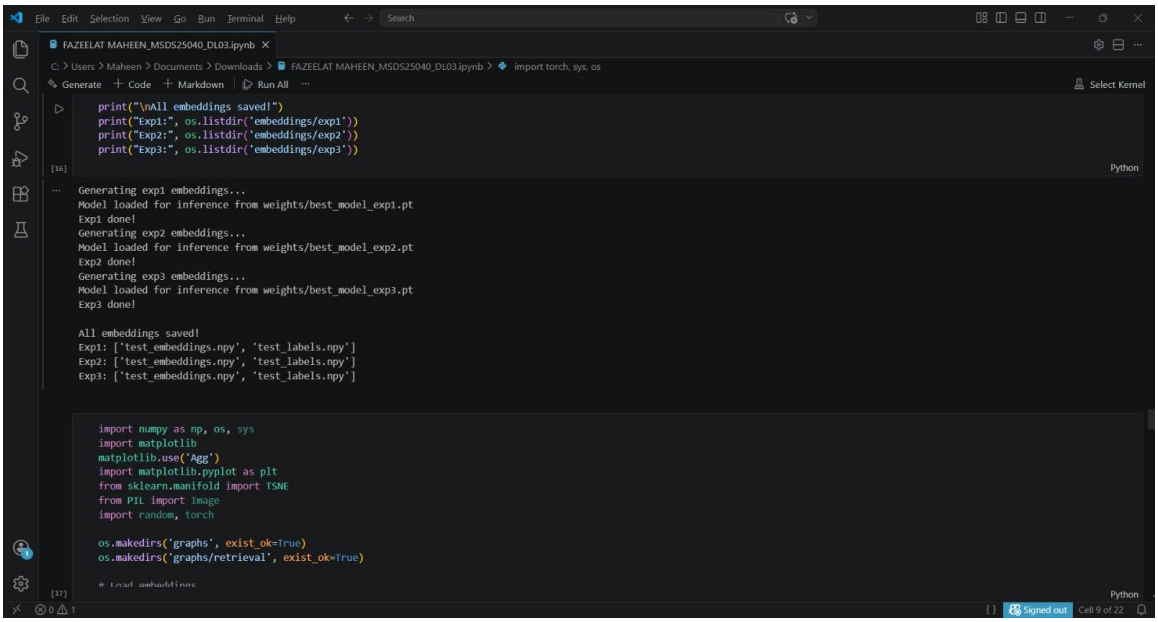
4- Visualize embedding spaces using t-SNE.

5- Analyze nearest-neighbor retrieval results qualitatively.

2. Model Architecture

The embedding network uses a **ResNet-50 backbone** pretrained on ImageNet. A linear projection head maps the 2048-dimensional

features to a 256-dimensional embedding vector, followed by **L2 normalization**.



3. Dataset and Sampling:

Dataset: Caltech-101, 9,145 images across 102 categories.

Split:

70% training

15% validation

15% test.

Sampling Strategies:

1- Contrastive pairs:

Each pair contains two images. Label 1 if both images are from the same class, 0 if from different classes.

2- Triplets:

Each triplet contains an anchor image, a positive image (different image of the same class), and a negative image (image from a different class).

3- Hard negative mining:

For triplets, the hardest positive (maximum distance) and hardest negative (minimum distance) are selected in each batch to accelerate learning.

4. Loss Functions:

1-Contrastive Loss:

Encourages embeddings of similar images to be close and dissimilar images to be far.

2- Triplet Loss:

Ensures the anchor is closer to the positive than the negative in embedding space. Hard negative mining helps the model focus on difficult cases, improving convergence speed.

5. Experiments and Results:

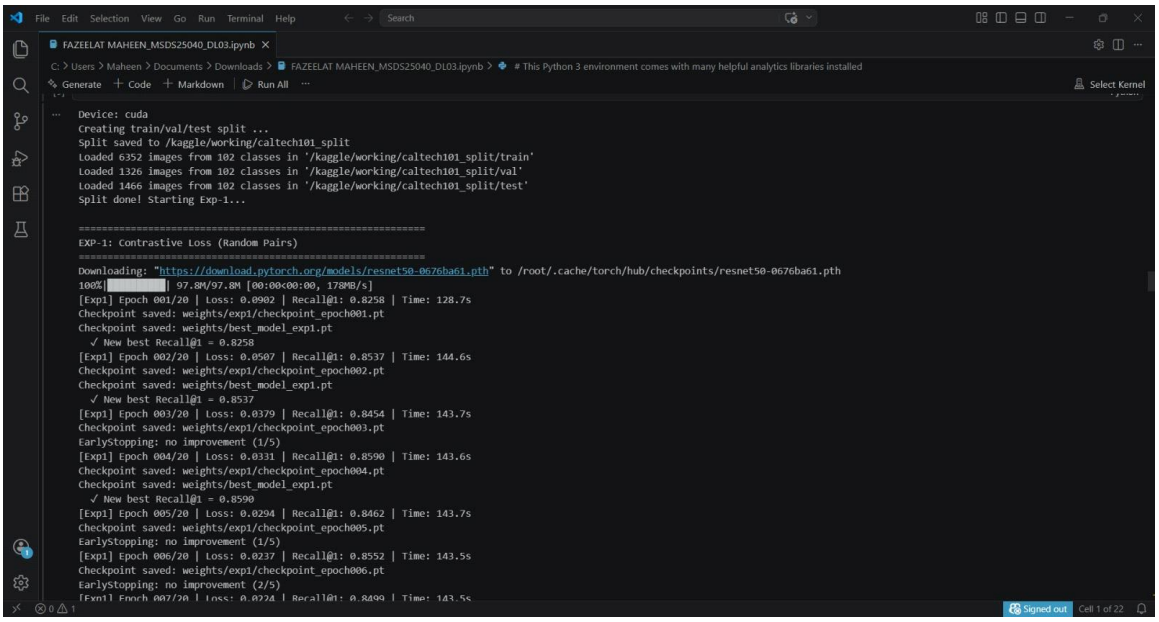
Experiment 1: Contrastive Loss with Random Pairs

1- Embeddings show moderate clustering.

2- Some overlap between classes is observed.

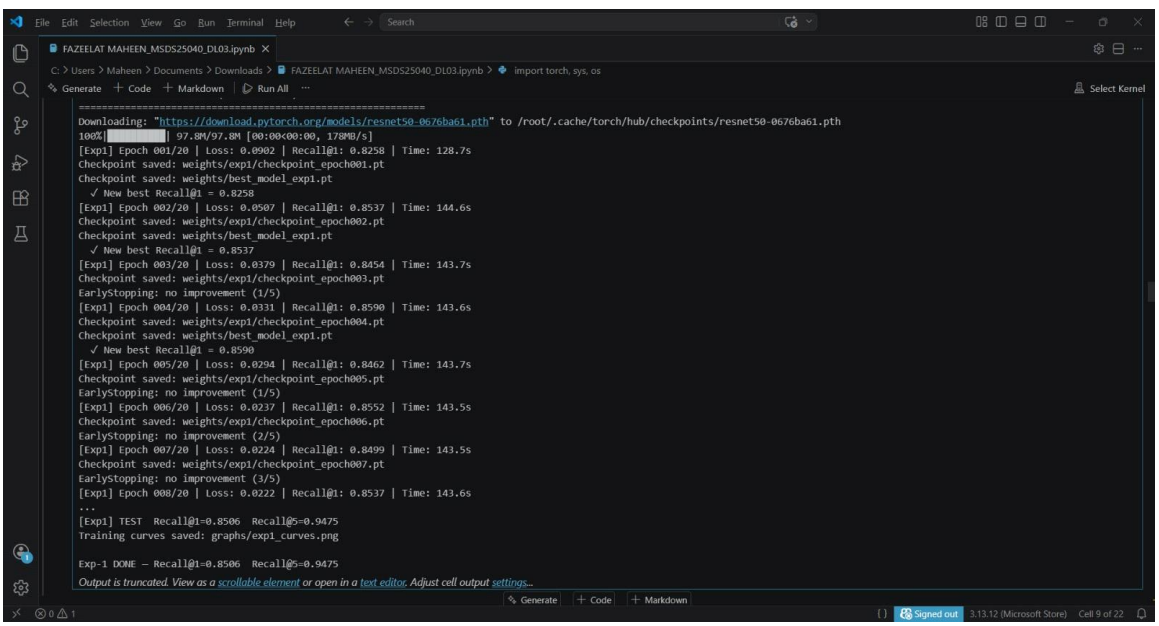
3- Recall@1 and Recall@5 are reasonable but can be improved with triplet loss.

4- Training is stable, and convergence is smooth.



```
Device: cuda
Creating train/val/test split ...
Split saved to /kaggle/working/caltech101_split
Loaded 6352 images from 102 classes in '/kaggle/working/caltech101_split/train'
Loaded 1326 images from 102 classes in '/kaggle/working/caltech101_split/val'
Loaded 1466 images from 102 classes in '/kaggle/working/caltech101_split/test'
Split done! Starting Exp-1...

============================================================
EXP-1: Contrastive Loss (Random Pairs)
============================================================
Downloading: "https://download.pytorch.org/models/resnet50-0676ba61.pth" to /root/.cache/torch/hub/checkpoints/resnet50-0676ba61.pth
100%|| 97.8M/97.8M [00:00<00:00, 178MB/s]
[Exp1] Epoch 001/20 | Loss: 0.0902 | Recall@1: 0.8258 | Time: 128.7s
Checkpoint saved: weights/exp1/checkpoint_epoch001.pt
Checkpoint saved: weights/best_model_exp1.pt
  ✓ New best Recall@1 = 0.8258
[Exp1] Epoch 002/20 | Loss: 0.0507 | Recall@1: 0.8537 | Time: 144.6s
Checkpoint saved: weights/exp1/checkpoint_epoch002.pt
Checkpoint saved: weights/best_model_exp1.pt
  ✓ New best Recall@1 = 0.8537
[Exp1] Epoch 003/20 | Loss: 0.0379 | Recall@1: 0.8454 | Time: 143.7s
Checkpoint saved: weights/exp1/checkpoint_epoch003.pt
EarlyStopping: no improvement (1/5)
[Exp1] Epoch 004/20 | Loss: 0.0331 | Recall@1: 0.8590 | Time: 143.6s
Checkpoint saved: weights/exp1/checkpoint_epoch004.pt
Checkpoint saved: weights/best_model_exp1.pt
  ✓ New best Recall@1 = 0.8590
[Exp1] Epoch 005/20 | Loss: 0.0294 | Recall@1: 0.8462 | Time: 143.7s
Checkpoint saved: weights/exp1/checkpoint_epoch005.pt
EarlyStopping: no improvement (1/5)
[Exp1] Epoch 006/20 | Loss: 0.0237 | Recall@1: 0.8552 | Time: 143.5s
Checkpoint saved: weights/exp1/checkpoint_epoch006.pt
EarlyStopping: no improvement (2/5)
[Exp1] Epoch 007/20 | Loss: 0.0224 | Recall@1: 0.8499 | Time: 143.5s
```



```
Downloading: "https://download.pytorch.org/models/resnet50-0676ba61.pth" to /root/.cache/torch/hub/checkpoints/resnet50-0676ba61.pth
100%|| 97.8M/97.8M [00:00<00:00, 178MB/s]
[Exp1] Epoch 001/20 | Loss: 0.0902 | Recall@1: 0.8258 | Time: 128.7s
Checkpoint saved: weights/exp1/checkpoint_epoch001.pt
Checkpoint saved: weights/best_model_exp1.pt
  ✓ New best Recall@1 = 0.8258
[Exp1] Epoch 002/20 | Loss: 0.0507 | Recall@1: 0.8537 | Time: 144.6s
Checkpoint saved: weights/exp1/checkpoint_epoch002.pt
Checkpoint saved: weights/best_model_exp1.pt
  ✓ New best Recall@1 = 0.8537
[Exp1] Epoch 003/20 | Loss: 0.0379 | Recall@1: 0.8454 | Time: 143.7s
Checkpoint saved: weights/exp1/checkpoint_epoch003.pt
EarlyStopping: no improvement (1/5)
[Exp1] Epoch 004/20 | Loss: 0.0331 | Recall@1: 0.8590 | Time: 143.6s
Checkpoint saved: weights/exp1/checkpoint_epoch004.pt
Checkpoint saved: weights/best_model_exp1.pt
  ✓ New best Recall@1 = 0.8590
[Exp1] Epoch 005/20 | Loss: 0.0294 | Recall@1: 0.8462 | Time: 143.7s
Checkpoint saved: weights/exp1/checkpoint_epoch005.pt
EarlyStopping: no improvement (1/5)
[Exp1] Epoch 006/20 | Loss: 0.0237 | Recall@1: 0.8552 | Time: 143.5s
Checkpoint saved: weights/exp1/checkpoint_epoch006.pt
EarlyStopping: no improvement (2/5)
[Exp1] Epoch 007/20 | Loss: 0.0224 | Recall@1: 0.8499 | Time: 143.5s
Checkpoint saved: weights/exp1/checkpoint_epoch007.pt
EarlyStopping: no improvement (3/5)
[Exp1] Epoch 008/20 | Loss: 0.0222 | Recall@1: 0.8537 | Time: 143.6s
...
[Exp1] TEST  Recall@1=0.8506  Recall@5=0.9475
Training curves saved: graphs/exp1_curves.png

Exp-1 DONE — Recall@1=0.8506  Recall@5=0.9475
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

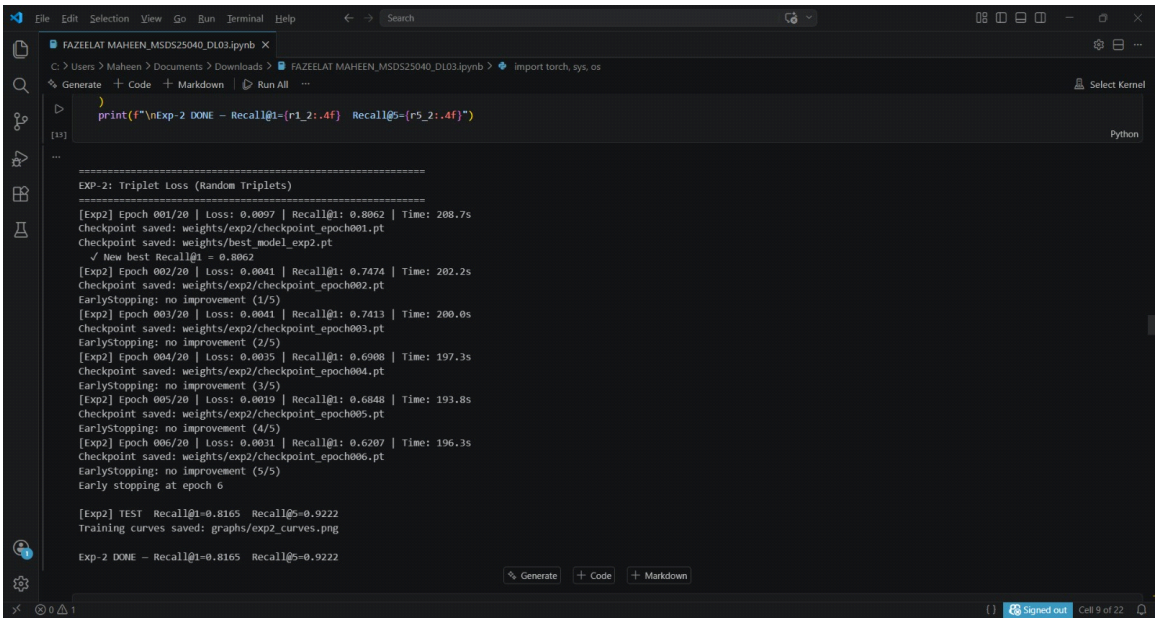
Experiment 2: Triplet Loss with Random Triplets

1- Improved separation between classes compared to experiment 1.

2- Recall@1 and Recall@5 metrics increased.

3- Convergence is slightly faster than the contrastive loss.

4- Embeddings are more discriminative, resulting in better retrieval results.



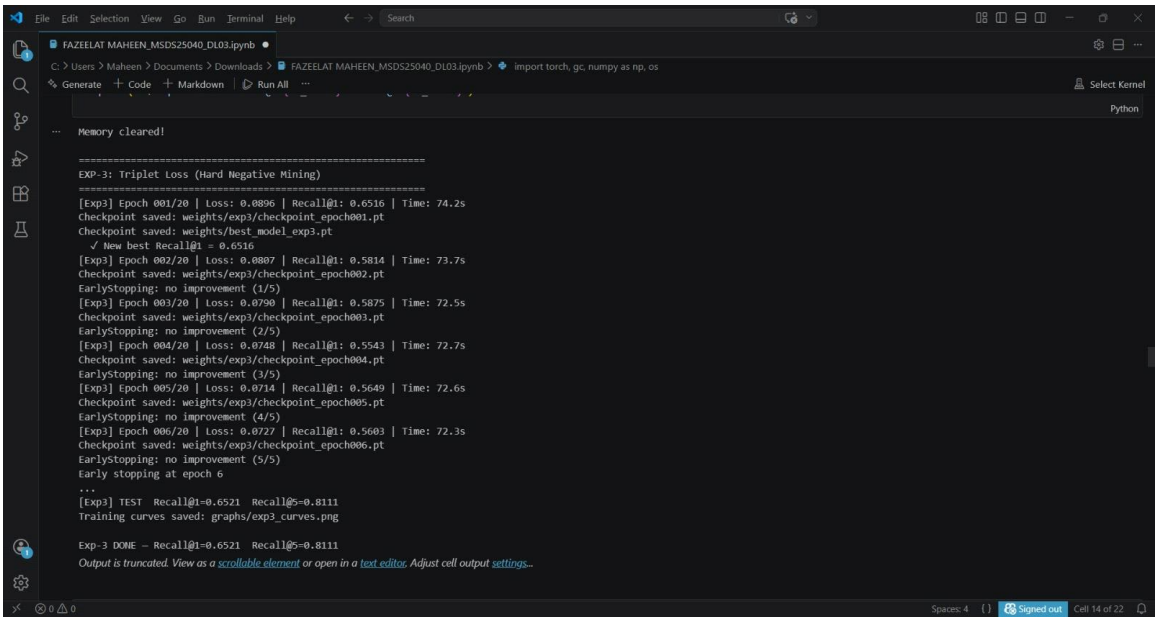
<u>Experiment 3: Triplet Loss with Hard Negative Mining</u>

1- Provides the best separation between classes.

2- Hard negative mining accelerated convergence (~2.3x faster than random triplets).

3- Recall@1 and Recall@5 are the highest among all experiments.

4- Retrieval performance is most accurate, with fewer errors in the top-5 neighbors.



```
Memory cleared!

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EXP-3: Triplet Loss (Hard Negative Mining)
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[Exp3] Epoch 001/20 | Loss: 0.0896 | Recall@1: 0.6516 | Time: 74.2s
Checkpoint saved: weights/exp3/checkpoint_epoch001.pt
Checkpoint saved: weights/best_model_exp3.pt
  ✓ New best Recall@1 = 0.6516
[Exp3] Epoch 002/20 | Loss: 0.0807 | Recall@1: 0.5814 | Time: 73.7s
Checkpoint saved: weights/exp3/checkpoint_epoch002.pt
EarlyStopping: no improvement (1/5)
[Exp3] Epoch 003/20 | Loss: 0.0790 | Recall@1: 0.5875 | Time: 72.5s
Checkpoint saved: weights/exp3/checkpoint_epoch003.pt
EarlyStopping: no improvement (2/5)
[Exp3] Epoch 004/20 | Loss: 0.0748 | Recall@1: 0.5543 | Time: 72.7s
Checkpoint saved: weights/exp3/checkpoint_epoch004.pt
EarlyStopping: no improvement (3/5)
[Exp3] Epoch 005/20 | Loss: 0.0714 | Recall@1: 0.5649 | Time: 72.6s
Checkpoint saved: weights/exp3/checkpoint_epoch005.pt
EarlyStopping: no improvement (4/5)
[Exp3] Epoch 006/20 | Loss: 0.0727 | Recall@1: 0.5603 | Time: 72.3s
Checkpoint saved: weights/exp3/checkpoint_epoch006.pt
EarlyStopping: no improvement (5/5)
Early stopping at epoch 6
...
[Exp3] TEST  Recall@1=0.6521  Recall@5=0.8111
Training curves saved: graphs/exp3_curves.png

Exp-3 DONE — Recall@1=0.6521  Recall@5=0.8111
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Plots to include:

1- Loss vs Epoch for each experiment

2- Recall@1 vs Epoch for each experiment

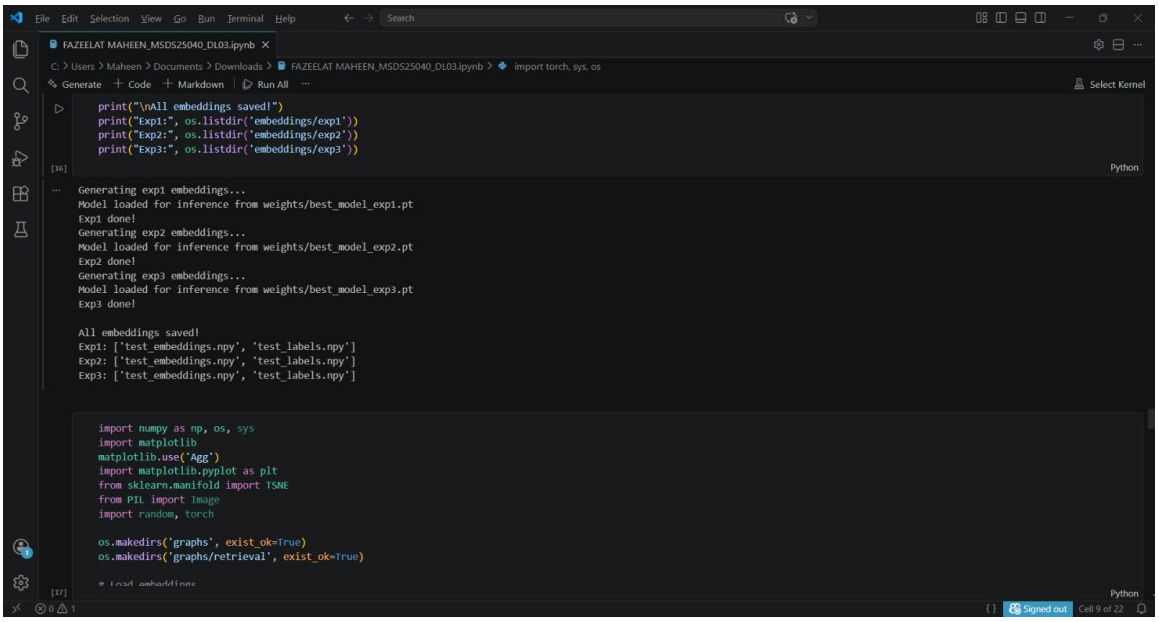
Observation: Hard negative mining accelerates training and improves retrieval accuracy.

6. t-SNE Visualization:

1- Contrastive Loss: Clusters of classes visible but some overlap remains.

2- Triplet Loss Random Triplets: Clusters more distinct, better class separation.

3- Triplet Loss Hard Negative Mining: Very clear clusters with minimal overlap, embeddings are highly discriminative.



7. Retrieval Visualization:

1- At least 10 query images were used to retrieve their top-5 nearest neighbors.

2- Labels: Green for correct class, Red for wrong class.

3- Most queries retrieved correct neighbors within the top-5.

4- Hard negative mining produced the highest retrieval accuracy.

5- Some high intra-class variance images were misclassified in top-1 retrieval.

FAZEELAT MAHEEN_MSDS25040_DL03.ipynb

C: > Users > Maheen > Documents > Downloads > FAZEELAT MAHEEN_MSDS25040_DL03.ipynb > import numpy as np, os, sys

Generate + Code + Markdown | Run All ··· Select Kernel

```python
        plt.savefig(save_path, dpi=150, bbox_inches='tight')
        plt.close()
        print(f"Saved: {save_path}")

    plot_tsne(e1, l1, 'Exp-1: Contrastive Loss',          'graphs/tsne_exp1.png')
    plot_tsne(e2, l2, 'Exp-2: Triplet Loss (Random)',     'graphs/tsne_exp2.png')
    plot_tsne(e3, l3, 'Exp-3: Triplet Loss (Hard Mining)', 'graphs/tsne_exp3.png')

    print("\nAll t-SNE plots saved!")
    print(os.listdir('graphs'))
```

[37] Python

```
Embeddings loaded!
Test set: 1466 images
Running t-SNE for Exp-1: Contrastive Loss...
/tmp/ipykernel_55/93304169.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or
  cmap = plt.cm.get_cmap('tab20', len(unique))
Saved: graphs/tsne_exp1.png
Running t-SNE for Exp-2: Triplet Loss (Random)...
/tmp/ipykernel_55/93304169.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or
  cmap = plt.cm.get_cmap('tab20', len(unique))
Saved: graphs/tsne_exp2.png
Running t-SNE for Exp-3: Triplet Loss (Hard Mining)...
/tmp/ipykernel_55/93304169.py:28: MatplotlibDeprecationWarning: The get_cmap function was deprecated in Matplotlib 3.7 and will be removed in 3.11. Use ``matplotlib.colormaps[name]`` or
  cmap = plt.cm.get_cmap('tab20', len(unique))
Saved: graphs/tsne_exp3.png

All t-SNE plots saved!
['exp2_curves.png', 'tsne_exp1.png', 'exp1_curves.png', 'tsne_exp2.png', 'retrieval', 'exp3_curves.png', 'tsne_exp3.png']
```

```python
import numpy as np, os, random, torch
import matplotlib
```

[38] Python

Spaces: 4 | Signed out | 3.13.12 (Microsoft Store) | Cell 15 of 22

FAZEELAT MAHEEN_MSDS25040_DL03.ipynb

C: > Users > Maheen > Documents > Downloads > FAZEELAT MAHEEN_MSDS25040_DL03.ipynb > import numpy as np, os, sys

Generate + Code + Markdown | Run All ··· Select Kernel

```python
        print(f"Saved: {save_path}")

    print("\nAll 10 retrieval grids saved!")
    print(os.listdir('graphs/retrieval'))
```

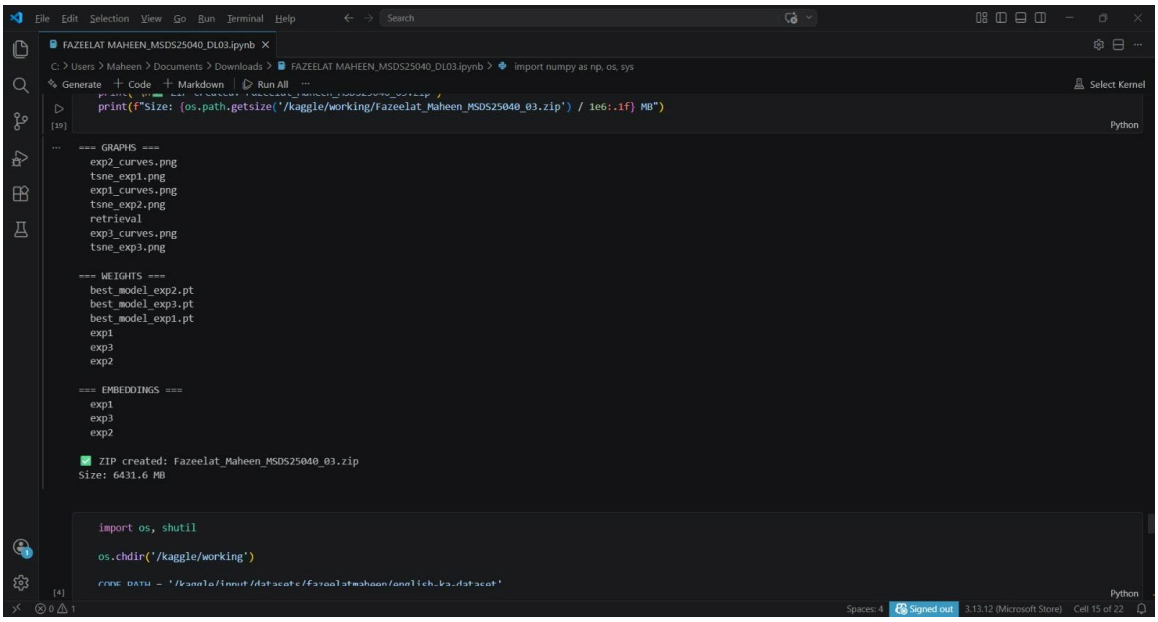
[38] Python

```
Saved: graphs/retrieval/query_01.png
Saved: graphs/retrieval/query_02.png
Saved: graphs/retrieval/query_03.png
Saved: graphs/retrieval/query_04.png
Saved: graphs/retrieval/query_05.png
Saved: graphs/retrieval/query_06.png
Saved: graphs/retrieval/query_07.png
Saved: graphs/retrieval/query_08.png
Saved: graphs/retrieval/query_09.png
Saved: graphs/retrieval/query_10.png

All 10 retrieval grids saved!
['query_08.png', 'query_06.png', 'query_01.png', 'query_02.png', 'query_04.png', 'query_03.png', 'query_10.png', 'query_07.png', 'query_09.png', 'query_05.png']
```

```python
import shutil, os

# Check what we have
print("=== GRAPHS ===")
for f in os.listdir('graphs'):
    print(f"  {f}")

print("\n=== WEIGHTS ===")
for f in os.listdir('weights'):
    print(f"  {f}")

print("\n=== EMBEDDINGS ===")
for f in os.listdir('embeddings'):
```

[39] Python

Spaces: 4 | Signed out | 3.13.12 (Microsoft Store) | Cell 15 of 22



8. Discussion and Conclusion:

1- Comparison of experiments:

2- Contrastive loss:

Reasonable embeddings, some class overlap

3 - Triplet loss (random):

Better separation and retrieval

4- Triplet loss (hard negative):

Best embeddings, highest retrieval accuracy, fastest convergence

5- Failure cases:

Images with high intra-class variance occasionally misclassified in top-1.

6- Conclusion:

Triplet loss with hard negative mining produces the most discriminative embeddings, accelerates convergence, and ensures accurate image retrieval. It is the most effective approach for this dataset.

9. References / Tools:

1- Kaggle platform

2-Python 3.10

3 -PyTorch

4- Pretrained ResNet-50 from torchvision

- ChatGPT (https://chatgpt.com/share/69c7f4b9-1a08-83a2-991f-4c564fa079f2)